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Lambda in natural language semantics: function application and lambda conversion: a web app

Available at: 'https://cbreul.org/cb-web/academ/lambda/lambda.pl'

The purpose of this little web app is to provide the output of an application of a function to its argument in the style of the lambda calculus, i.e. in the form *λx [P (x)] (y)*, where the part *λx [P (x)]* is the function and *y* is the argument that the function is applied to. The page is geared to uses of the lambda calculus in natural language semantics, as introduced in Kearns (2000/2011: chap. 4) and Lohnstein (1996/2011: chap. 8), for instance. Examples of function applications that can be processed by the app and their results are as follows. (The examples partially differ slighly in notational style; the actual form of the input in the app requires the use of an underscore at certain places for computational reasons.)

λx [SHOW (Jo, x, Chris)] (the book) ➔ SHOW (Jo, the book, Chris)

λx [λy [(study (x)) (y)]] (‖linguistics‖) ➔ λy [(study (‖linguistics‖)) (y)]

λP [ιx (P (x))] (λy [cat' (y)]) ➔ ιx (λy [cat' (y)] (x))

λy[treu'(y) ∧ Gefährte'(y)] (p) ➔ treu'(p) ∧ Gefährte'(p)

Φ = λx [sleep (x)] (Jo) ∧ Tsit (s) ⊂ Tsit (Φ) ➔ Φ = sleep (Jo) ∧ Tsit (s) ⊂ Tsit (Φ)

λQ [λP [∀x [Q (x) → P (x)]]] (λy [Pflanze' (y)]) ➔ λP [∀x [λy [Pflanze' (y)] (x) → P (x)]]

The app may be helpful in the context of teaching the lambda calculus in semantics classes. As input and output can be copied and pasted, it may also be useful for saving time and effort for typing.

References:

Kearns, Kate. 2000/2011. Semantics. 2nd edn. Houndmills: Palgrave Macmillan.

Lohnstein, Horst. 1996/2011. Formale Semantik und natürliche Sprache. 2nd edn. Berlin: De Gruyter.